www.ddr2.org
The Web site for DDR2 news, technology updates, white papers, press releases, etc.

http://www.seagate.com/newsinfo/technology/research/D4b1.html
A Web site about the Seagate Research centre, and its activities; Seagate is at the cutting edge of storage technologies.

http://www.storagesearch.com/ssd.html
News about solid-state storage devices; includes articles and links to company Web sites.

http://www.pcguide.com/ref/hdd/op/heads/
An interesting Web site about hard disk heads; Contains both basic as well as advanced information.

http://uk.computers.toshiba-europe.com/cgi-bin/ToshibaCSG/future_techs. jsp?z=64&service=UK
This page has links to lots of storage technology articles, from RAID to DVD and emerging technologies.

http://www.inphase-technologies.com/technology/
Read about holographic storage technology in detail.

several university and company research labs are working on IRAM architectures.

MRAM

Magnetic Random Access Memory (MRAM) is non-volatile, like Flash. It holds promise because, when designs are complete, it can combine the high speeds of SRAM, the capacities of DRAM, and the non-volatile nature of Flash. In an MRAM chip, the memory cells are the intersections of the rows and columns. Electric signals polarise the cells into ones and zeroes. MRAM is like a Flash stick—there are no moving parts; besides, power consumption is lower. It's therefore likely that MRAM and Flash will compete in the portable-device arena. MRAM has the edge because it will prolong battery life. As mainstream storage, MRAM chips are unlikely to hit the market anytime soon. IBM, however, is working on a combination of MRAM and holographic memory.

Looking forth

With so many materials and techniques being explored, one can't assume that storage will always be magnetic disks, or that memory will always be dynamic RAM. Systems of the future may build upon the concepts that exist today, or they may employ entirely new technologies such as holographic memory, or they may even bring back old ideas. Is tape the storage of the future? It is, according to Pat Martin, CEO of StorageTek! Predictions vary.

But we can be sure of a few things—for example, perpendicular recording will happen very soon; MRAM will be researched more aggressively, because of its promise as a replacement for Flash as well as becoming a PC RAM solution, etc.

Whatever happens in the future, we're quite far off from the day you'll be able to buy a computing system and live with it happily ever after—sans upgrades.

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